



Von Neumann architecture



Is based on three key concepts:

- Data and instructions are stored in a single read-write memory.
- The contents of this memory are addressable by location .
- Execution occurs in a sequential fashion from one instruction to the next.



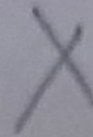
Computer Components



- Computer consists of **CPU** (central processing unit), memory, and **I/O** components, with one or more modules of each type.
- These components are interconnected in some fashion to achieve the basic function of the computer, which is to execute programs.



Computer Components



we can characterize a computer system by describing :

- (1) - The external behavior of each component, that is, the data and control signals that it exchanges with other components .
- (2) - The interconnection structure and the controls required to manage the use of the interconnection structure.



INTERCONNECTION STRUCTURES



- ❑ **Memory:** a memory module will consist of N words of equal length . each word is assigned a unique numerical address $(0, 1, \dots, N - 1)$. A word of data can be read from or written into the memory .The nature of the operation is indicated by read and write control signals.
- ❑ **I/O module:** I/O is functionally similar to memory. There are two operations, read and write . I/O module may control more than one external device. We can refer to each of the interfaces to an external device as a port and give each a unique address (e.g., $0, 1, \dots, M - 1$).

BUS INTERCONNECTION

- ☐ Data lines - provide a path for moving data among system modules. These lines, collectively, are called the data bus. The **data bus** may consist of 32, 64, 128, or even more separate lines
- ☒ For example : if the data bus is 32 bits wide and each instruction is 64 bits long, then the processor must access the memory module twice during each instruction cycle.

BUS INTERCONNECTION

□ Address lines are used to designate the source or destination of the data on the data bus.

For example, if the processor wishes to read a word (8, 16, or 32 bits) of data from memory, it puts the address of the desired word on the address lines.

The address lines are generally also used to **address I/O ports**.

BUS INTERCONNECTION

- Control lines : are used to control the access to and the use of the data and address lines.
- ✗ Because the data and address lines are shared by all components, there must be a means of controlling their use.

Control signals transmit both command and timing information among system modules. Timing signals indicate the validity of data and address information. Command signals specify operations to be performed.

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BUS INTERCONNECTION

Typical control lines include :

- **Memory write:** Causes data on the bus to be written into the addressed location.
- **Memory read:** Causes data from the addressed location to be placed on the bus .
- **I/O write:** Causes data on the bus to be output to the addressed I/O port
- **I/O read:** Causes data from the addressed I/O port to be placed on the bus .
- **Transfer ACK:** Indicates that data have been accepted from or placed on the bus .

BUS INTERCONNECTION

Typical control lines include :

- **Bus request:** Indicates that a module needs to gain control of the bus .
- **Bus grant:** Indicates that a requesting module has been granted control of the bus .
- **Interrupt request:** Indicates that an interrupt is pending.
- **Interrupt ACK :** Acknowledges that the pending interrupt has been recognized.
- **Clock:** Is used to synchronize operations .
- **Reset:** Initializes all modules .

BUS INTERCONNECTION

- The operation of the bus is as follows. If one module wishes to send data to another, it must do two things:
 - 1- obtain the use of the bus .
 - 2- transfer data via the bus.

If one module wishes to request data from another module, it must :

- 1- obtain the use of the bus .
- 2- transfer a request to the other module over the appropriate control and address lines. It must then wait for that second module to send the data.